JLR

RANGE ROVER DEFENDER DISCOVERY

Emergency Response Guide

Version: 1
Dated: 12/12/2025

RANGE ROVER DEFENDER DISCOVERY

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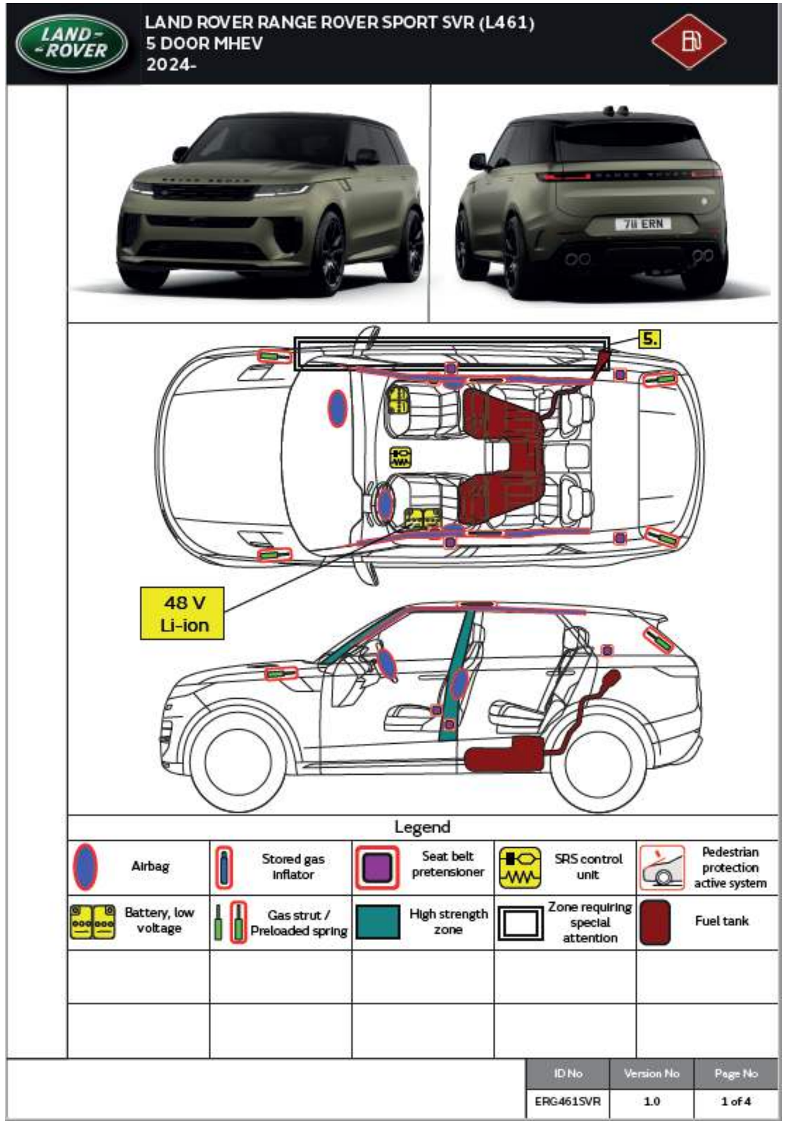
0. Rescue Sheet(s)

Rescue Sheet and Emergency Response Guide site.
JLR produce rescue sheets for all their vehicles, they can be found on the following link:

https://www.ownerinfo.landrover.com/rescue



Vehicle applicability.
The Emergency Response Guide applies to Land Rover vehicles produced by JLR, specifically the ones shown in the following table.

All vehicles or variants may not be available in every market.

Current Model Range

	**Range Rover Evoque (L551)** 2019MY - Onwards Petrol Diesel Hybrid (PHEV & MHEV)
	**Land Rover Discovery (L462)** 2017MY - Onwards Petrol Diesel Hybrid (MHEV)
	**Land Rover Discovery Sport (L550)** 2015MY - Onwards Petrol Diesel Hybrid (PHEV & MHEV)

Current Model Range

	**Range Rover (L460)** 2022MY - Onwards Petrol Diesel Hybrid (PHEV & MHEV)
	**Range Rover Sport (L461)** 2023MY - Onwards Petrol Diesel Hybrid (PHEV & MHEV)
	**Range Rover Velar (L560)** 2018MY - Onwards Petrol Diesel Hybrid (PHEV & MHEV)
	**Land Rover Defender 90, 100 & 130 (L663)** 2018MY - Onwards Petrol Diesel Hybrid (PHEV & MHEV)

1. Identification / recognition

⚠	Lack of engine noise does not mean vehicle is shut off: Silent movement or instant restart capability exists until the vehicle is fully shut down. Wear appropriate PPE.

Identification of Land Rover models.
Land Rover vehicles can be identified through the following badging.

Some models do not display vehicle name badges (4), for example Range Rover (L460) and Range Rover Sport (L461).

*Not applie on all models

ID	Description	Examples
1	Brand badge, front grille	LAND ROVER
2	Bonnet script	RANGE ROVER
3	Tailgate script	RANGE ROVER
4*	Vehicle name badge	VELAR EVOQUE SPORT
5*	Brande badge, tailgate	LAND ROVER

Land Rover powertrain variants.
Land Rover offers different powertrain options, this includes:

- Internal Combustion Engine (ICE)
- Mild-Hybrid Electric Vehicles (MHEV)
- Plug-in Hybrid Electric Vehicles (PHEV)

These powertrain options feature electrical systems which operate at varying voltages and therefore present different hazards to first responders.

More information about the battery chemistry, voltages and locations can be found in section 5.

Key.

- Fuel.

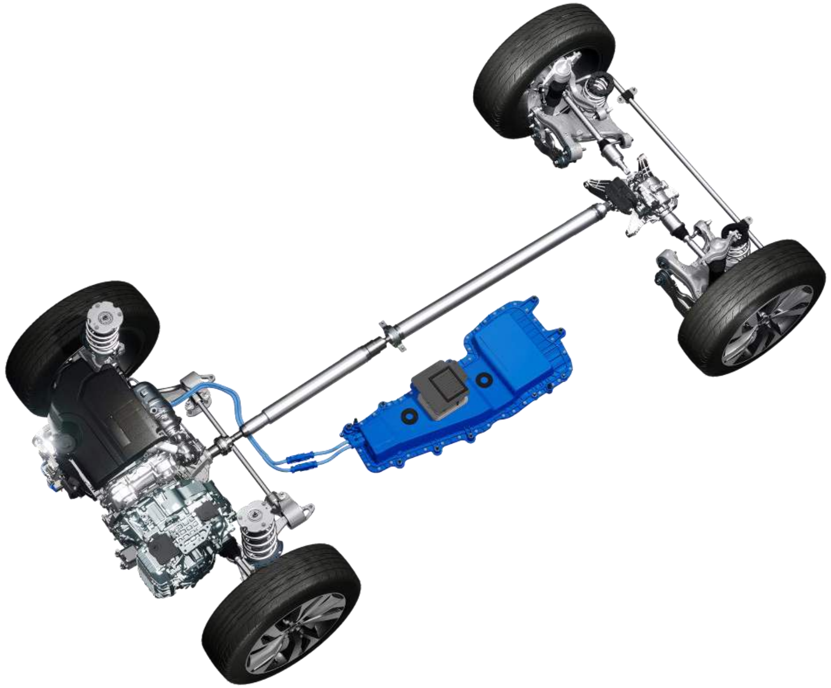
- Low voltage battery (48v).

- High voltage battery.

	ICE	MHEV	PHEV
Energy source			
Voltage	12 v	12 v – 48 v	12 v – 400 v
Electric driver motor	-	22 kW	160 kW
Electric driving range	-	-	40 – 91 km

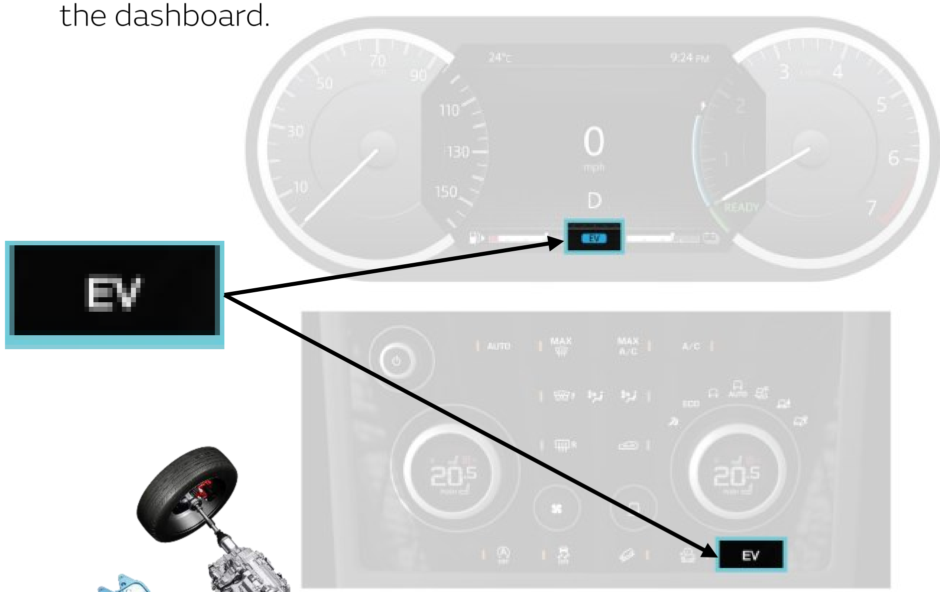
Identification of Land Rover variants.
Land Rover models come in different variants including ICE, MHEV, and PHEV.
It may not always be possible to determine the variant of the vehicle.

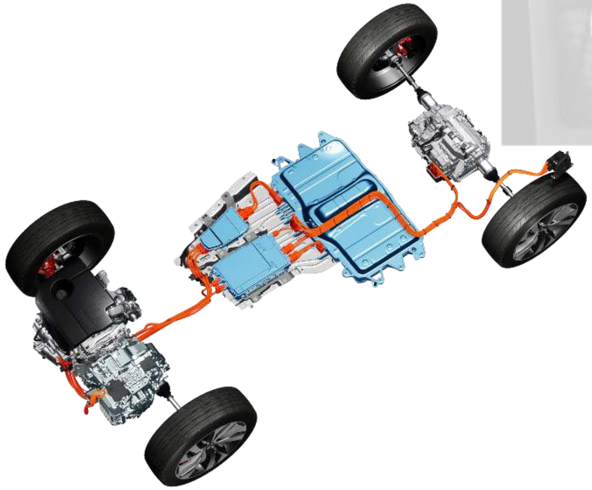
ICE/MHEV.
It may not be possible to distinguish between ICE and MHEV vehicles. An MHEV vehicle will have blue es, these can be seen around the MHEV battery location and sometimes within the engine bay.



PHEV.
PHEV vehicles can be identified by the following:

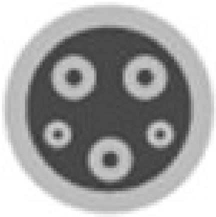
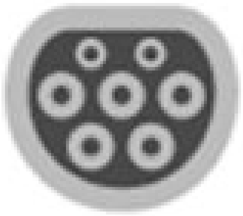
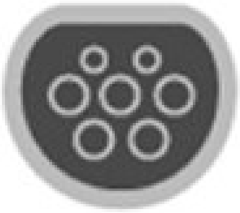
1. Orange HV es.
2. Charge port on the left side of the vehicle.
3. An “EV” symbol in the instrument cluster and a button on the dashboard.

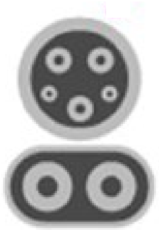
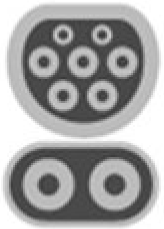
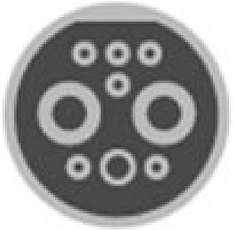




High Voltage charging socket.
Land Rover vehicles use different Alternating Current (A/C) and Direct Current (D/C) charging socket styles depending on the market.

Vehicles can have both an A/C socket and D/C socket installed on the vehicle, typically on opposite sides. Care must be taken to correctly identify.

	A/C	North America Canada Japan
	A/C	United Kingdom Europe Rest of the World
	A/C	China

	D/C	Japan
	D/C	North America Canada Korea
	D/C	United Kingdom Europe Rest of the World
	D/C	China

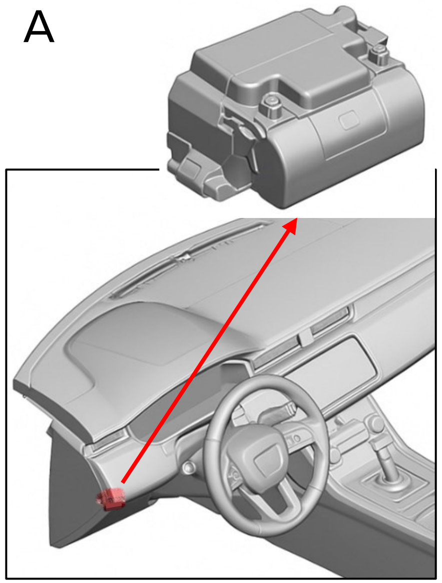
2. Immobilization / stabilization / lifting

Lifting Land Rover vehicles.
Prevent the vehicle from rolling away.
Before lifting the vehicle, it must be prevented from rolling away, where possible a block should be placed at the wheels to prevent rolling.

The Electronic Park Brake (EPB) is applied automatically when the vehicle's ignition is switched off, and the vehicle's speed is below 3 km/h (2 mph). The transmission also engages P to further secure the vehicle.

The (EPB) is located next to the steering wheel (A) or integrated with the Park (P) button (B) located on the centre console.

A



EPB location example – Next to the steering wheel.

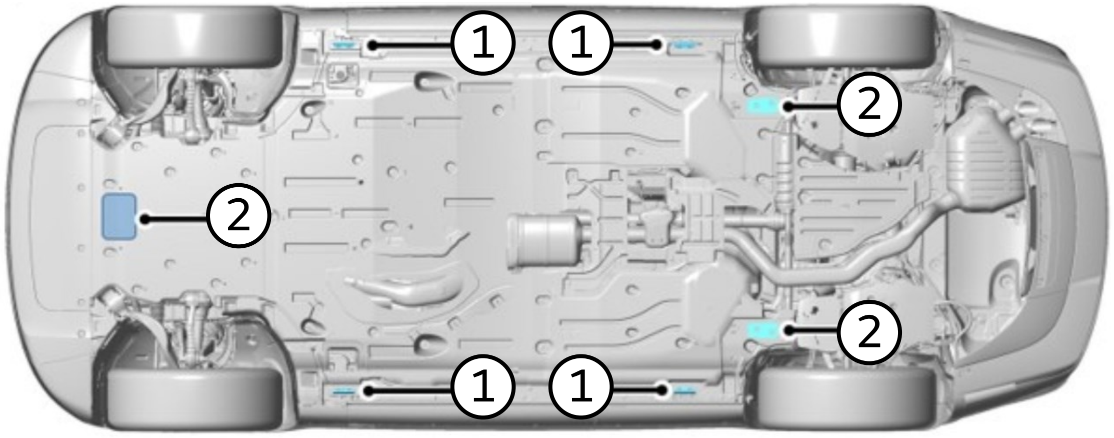
B

EPB location example – Integrated in the Park button.

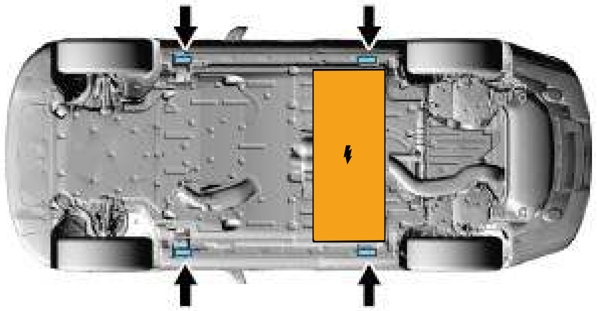
Lifting Points.
All Land Rover vehicles feature primary lifting points on the sills (1) and secondary lifting points underneath the vehicles at the front and rear (2).

The lifting points can be used for lifting equipment and axle stands.

The points for each vehicle are shown in the following pages.

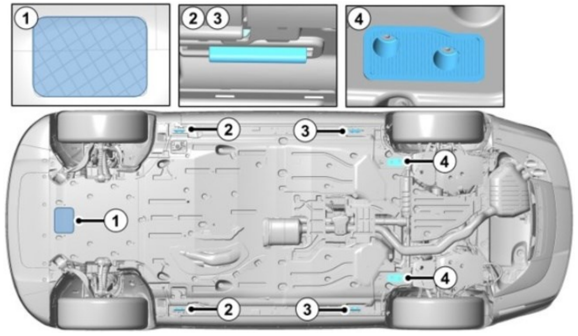


⚠	If the vehicle has been involved in an accident the First responders must assess the vehicle for safety before lifting.



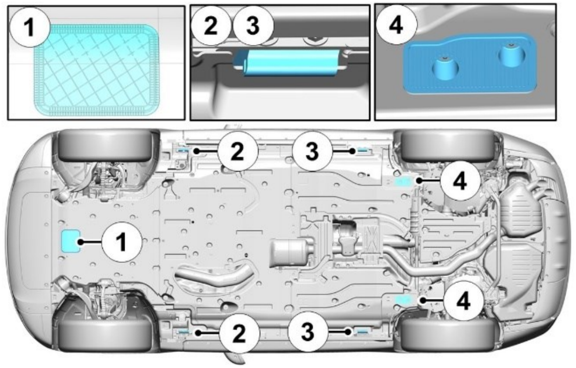
Care must be taken on vehicles fitted with a High Voltage battery (PHEV). The vehicle must not be lifted on the battery. Refer to the specific Rescue Sheet for battery location.

Range Rover (L460)



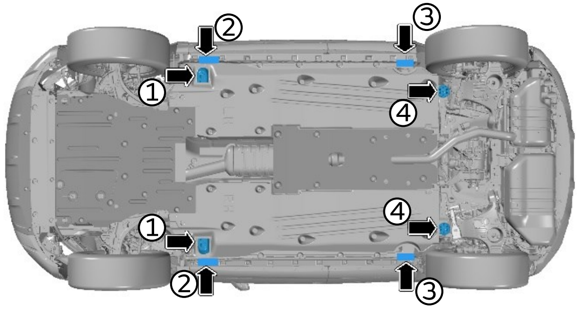
1. Front subframe lifting point.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

Range Rover Sport (L461)



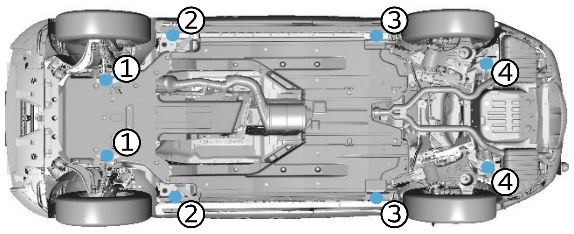
1. Front subframe lifting point.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

Range Rover Evoque (L551)



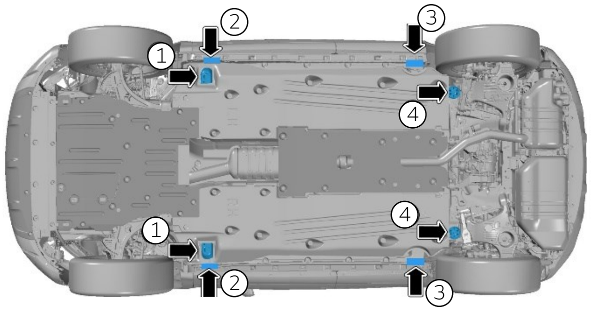
1. Front subframe lifting points.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

Range Rover Velar (L560)



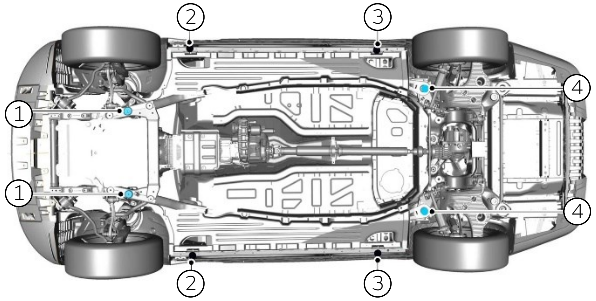
1. Front subframe lifting points.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

Land Rover Discovery Sport (L550)



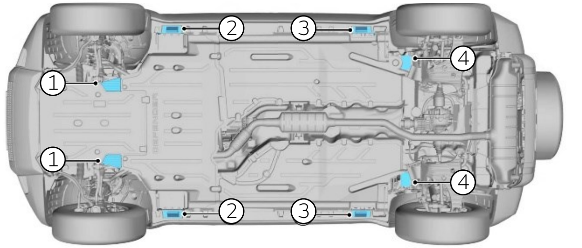
1. Front subframe lifting points.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

Land Rover Discovery (L462)



1. Front subframe lifting points.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

Land Rover Defender (L663)



1. Front subframe lifting points.
2. Front sill lifting points.
3. Rear sill lifting points.
4. Rear subframe lifting points.

3. Disable direct hazards / safety regulations

Hazard Disabling - General information.
On all Land Rover vehicles, disconnecting the 12 V system will isolate all electrical systems including the 48 V and High Voltage circuits.

Some vehicles will be equipped with two 12 V batteries, these are highlighted further in Section 5, please also refer to the relevant Rescue Sheet for specific information and locations.

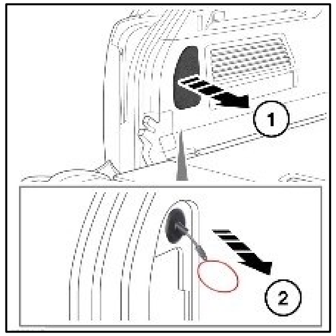
Alternative HV isolation methods are installed on some Land Rover vehicles; this will be highlighted within this section and the relevant Rescue Sheets.

⚠	The high voltage system is automatically disabled if the airbags are deployed.

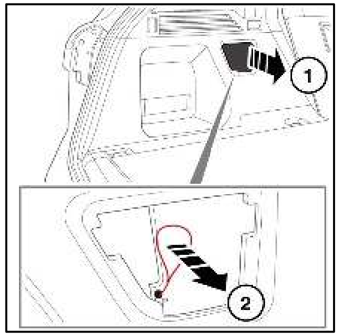
External charging.
Charging stations operate with either AC, DC or both, and they can be connected to a national grid with voltages over 1,000 V, special care needs to be taken when disconnecting in the event of an accident.

All external charging should be disconnected before attempting to disable the vehicle.
If it is not possible to release the charging e, vehicles have an emergency release e which can be found in the loadspace area. Images of the location of all HV vehicles are shown here.

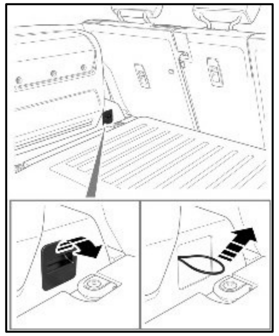
⚠	Incorrect handling of high voltage components poses a risk to life.

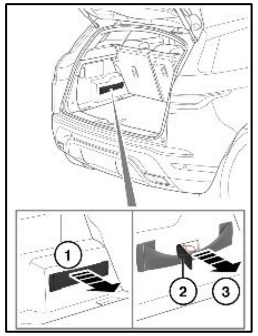


Range Rover (L460)

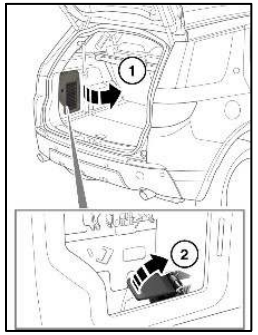


Range Rover Sport SVR (L461)

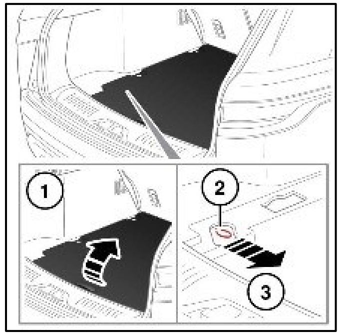

Land Rover Defender (L663)



Range Rover Evoque(L551)



Land Rover Discovery Sport (L550)



Range Rover Velar (L560)

12 V system disconnect.
The negative terminal should always be removed first before the positive terminal.
Once the terminals are disconnected the es should be prevented from accidently re-connecting or touching the terminal post.

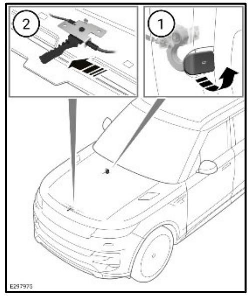
Some vehicles will be installed with two 12 V batteries, please refer to the relevant Rescue Sheet for specific information and location. The location of the batteries are also shown in Section 5.

NOTE: Disconnecting the 12 V system will also isolate the 48 V and High Voltage systems.

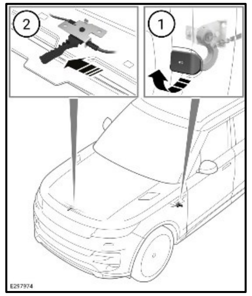
Alternative High Voltage System Isolation.
The Range Rover (L460) and Range Rover Sport (L461) both feature a cut-loop which can be used as an alternative high voltage isolation method. The cut-loop operates on a low voltage (12 V) so there is no exposure to high voltage using this method.

To access the cut-loop, the hood will need to be opened. To do this, operate the lever inside the vehicle then operate the lever under the hood. The interior lever location depends if the vehicle is left-hand or right-hand drive. To isolate the high voltage system, the wire should be cut in 2 places, one either side of the label. Once the e is cut, the high voltage system will be completely disabled.

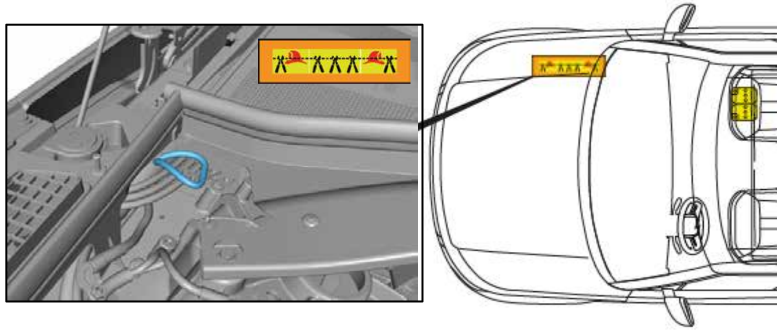
NOTE: Once the cut-loop is cut, the high voltage system cannot be started again.



Hood lever locations example – Range Rover Sport (L461) ***RHD***



Hood lever locations example – Range Rover Sport (L461) ***LHD***

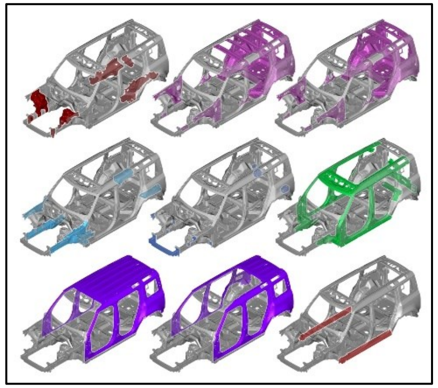


⚠	Cutting the cut-loop will only isolate the high voltage system, it will not isolate the low voltage system(s).

4. Access to the occupants

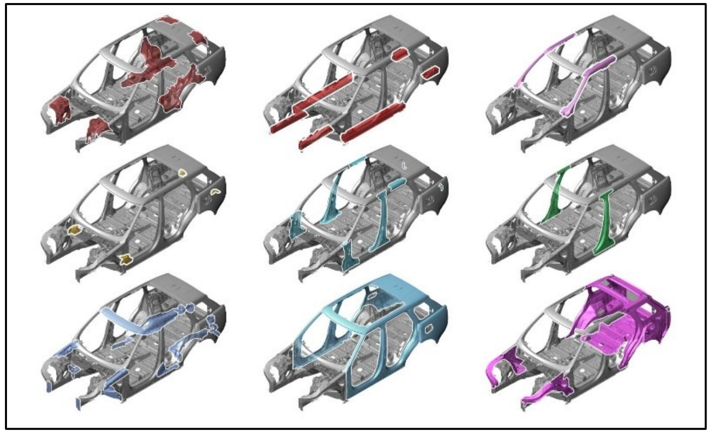
Vehicle body – Shell material.
Land Rover vehicles are constructed predominantly from steel and aluminum; the specific information is shown below.

Range Rover Defender (L663).
The Range Rover Defender shell is constructed fully from aluminum.



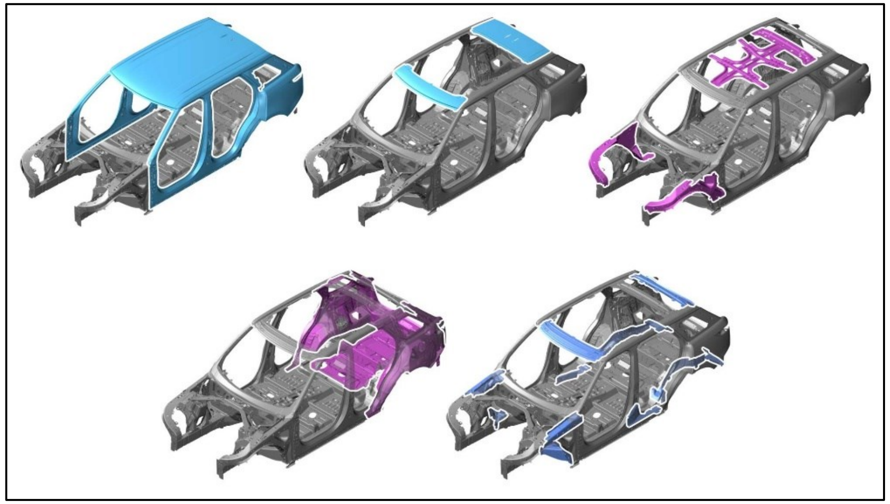
	Material	Type
	Aluminum	High Pressure Die Cast (HPDC)
	Aluminum	5 Series
	Aluminum	AC Series (6 Series)
	Aluminum	6 Series
	Aluminum	A Series (6 Series)
	Aluminum	AC Series (6 Series)
	Aluminum	Extrusion

Range Rover (L460).
The Range Rover shell is constructed from a combination of steel and aluminum.

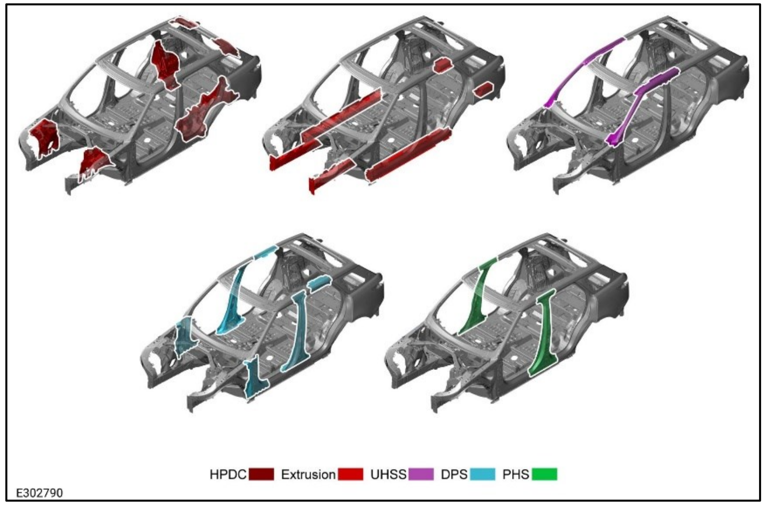


	Material	Type
	Aluminum	High Pressure Die Cast
	Aluminum	Extrusion
	Steel	Ultra High Strength Steel
	Steel	Mild Steel
	Steel	Dual Phase
	Steel	Press Hardened Steel
	Aluminum	6-Series
	Aluminum	AC-Series (6-Series)
	Aluminum	5-Series

Range Rover Sport (L461).
The Range Rover shell is constructed from a combination of steel and aluminum.



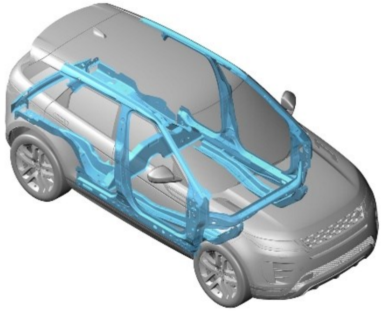
	Material	Type
	Aluminum	AC Series (6 Series)
	Aluminum	5 Series
	Aluminum	6 Series



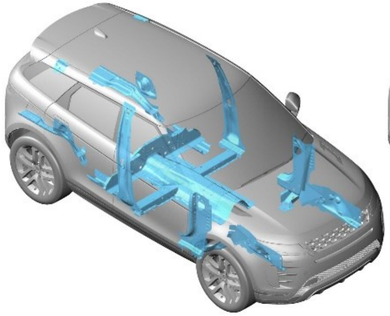
	Material	Type
	Aluminum	High Pressure Die Cast
	Aluminum	Extrusion
	Steel	Ultra High Strength Steel
	Steel	Dual Phase
	Steel	Press Hardened Steel

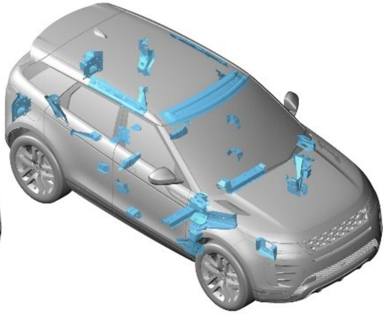
Range Rover Evoque (L551).
The Range Rover Evoque shell is constructed from a combination of steel and aluminum.

Ultra High Strength Steel (UHSS).

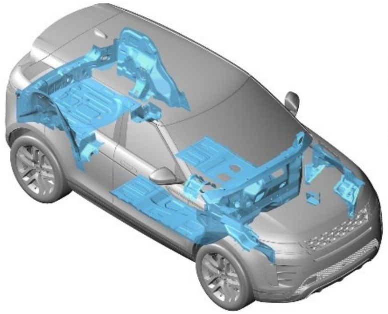


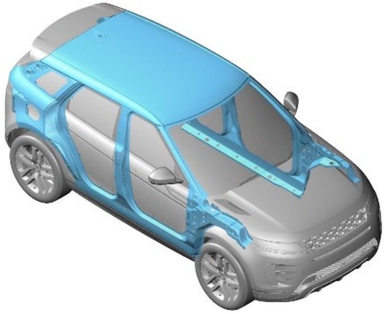
High Strength Steel (HSS).



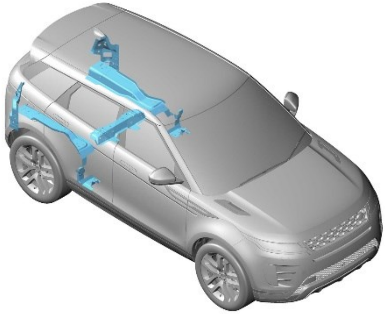


Mild Steel.

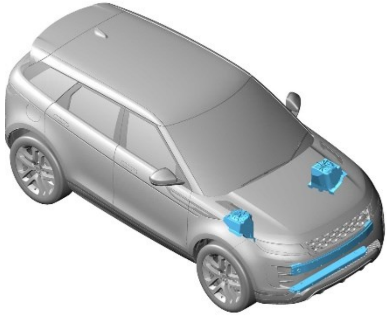




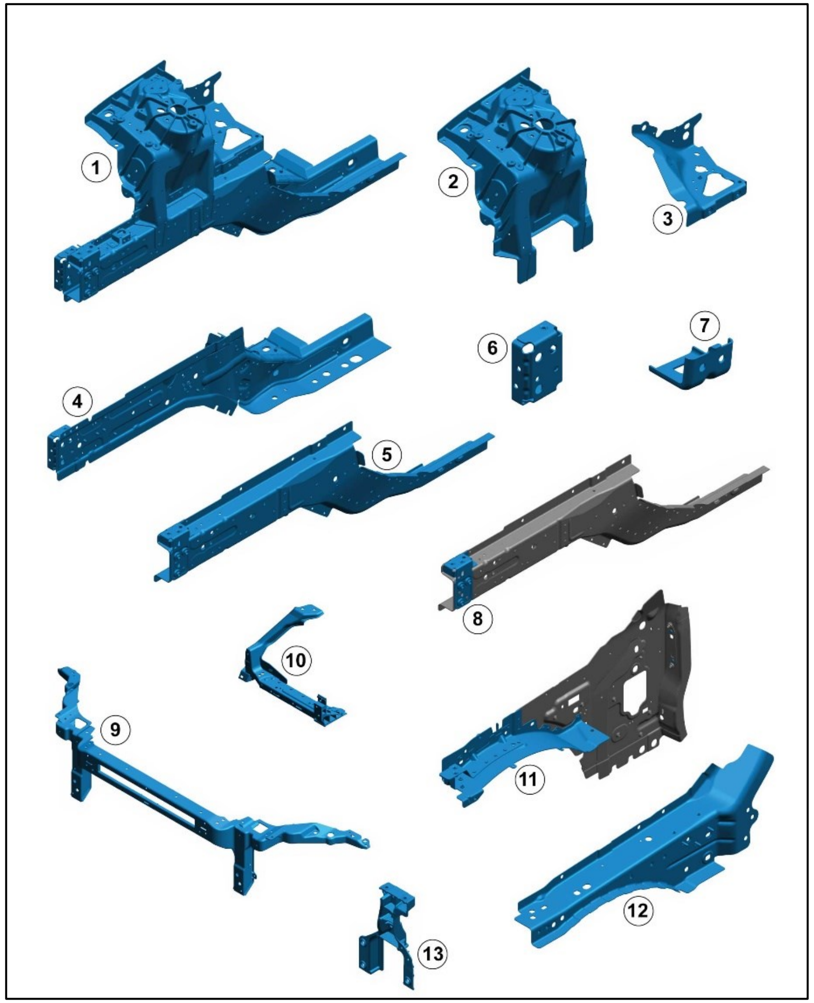
Bake Hardened (BH) Steel.



Aluminum.

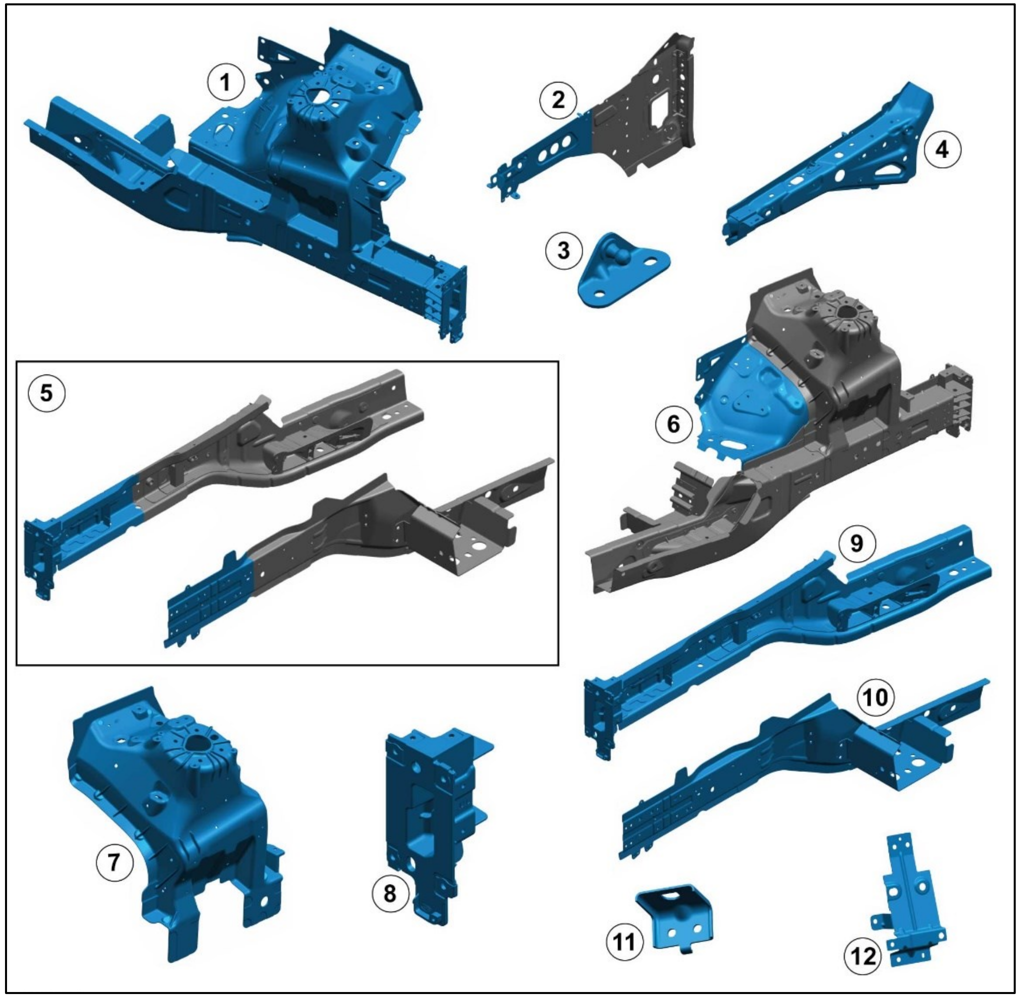


Range Rover Velar (L560).



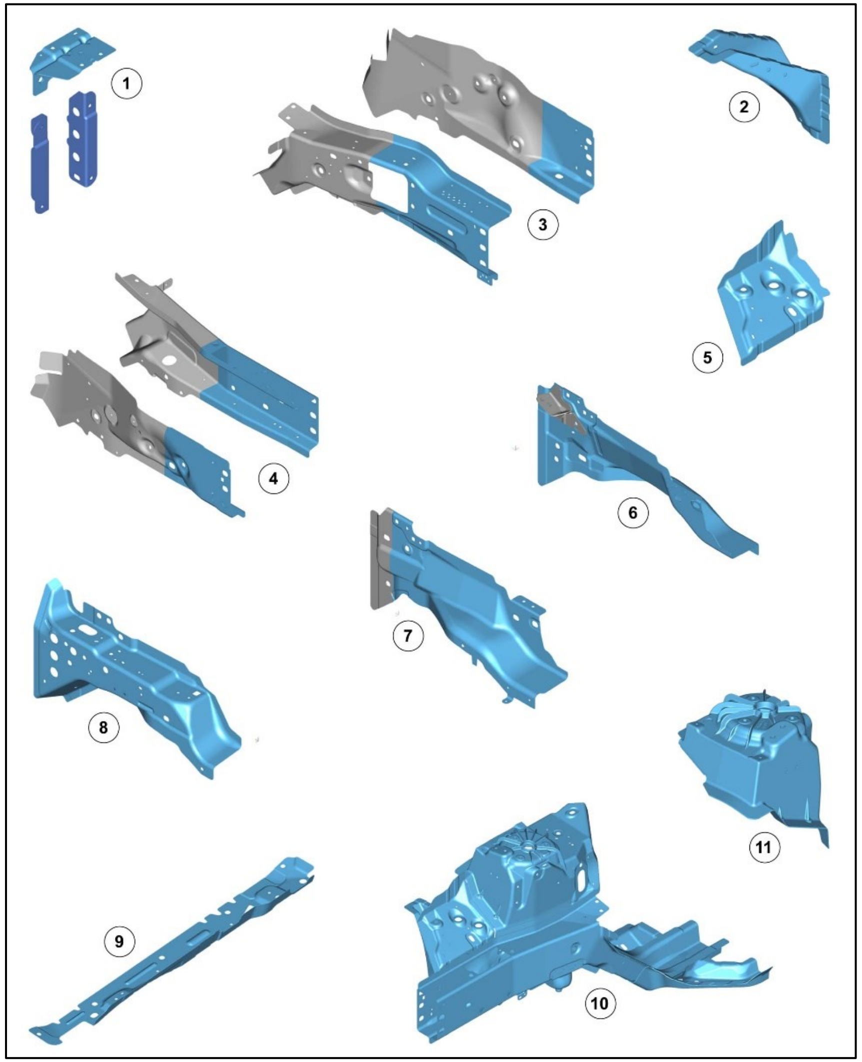
	Material	Type
1	Front side member and suspension top mount assembly	High Pressure Die Cast (HPDC) Aluminum + 5 Series Aluminum + 6 Series Aluminum
2	Suspension top mount	High Pressure Die Cast (HPDC) Aluminum
3	Front wheelhouse	5 Series Aluminum
4	Front side member closing panel	6 Series Aluminum
5	Front side member	6 Series Aluminum
6	Headlamp Mounting Support Bracket	6 Series Aluminum
7	Brake Hose Support Bracket	5 Series Aluminum
8	Hood Latch Panel Mounting Bracket	6 Series Aluminum
9	Hood latch panel	Magnesium Die Cast Alloy
10	Headlamp mounting bracket	High Pressure Die Cast (HPDC) Aluminum
11	Fender apron panel reinforcement	5 Series Aluminum
12	Fender apron panel	5 Series Aluminum
13	Front crossmember support bracket	High Pressure Die Cast (HPDC) Aluminum

Range Rover Discover (L462).



	Material	Type
1	Front Side Member and Suspension Top Mount Assembly	6 Series Aluminum + High Pressure Die Cast (HPDC) Aluminum
2	Fender Apron Panel Closing Panel	5 Series Aluminum
3	Hood Strut Bracket	High Strength Low Alloy (HSLA) Steel
4	Fender Apron Panel	5 Series Aluminum
5	Front Side Member Section	6 Series Aluminum + HPDC Aluminum
6	Front Wheelhouse	5 Series Aluminum
7	Suspension Top Mount	HPDC Aluminum
8	Front Bumper Mounting	HPDC Aluminum
9	Front Side Member	6 Series Aluminum + HPDC Aluminum
10	Front Side Member Closing Panel	6 Series Aluminum
11	Brake Hose Support Bracket	5 Series Aluminum
12	Front Fender Mounting Bracket Lower	5 Series Aluminum

Land Rover Discover Sport (L550).



	Material	Type
1	Front Bumper Mounting	Steel
2	Fender Apron Lower Reinforcement	Steel
3	Front Side Member Section Left Side	High Strength Steel
4	Front Side Member Section Right Side	High Strength Steel
5	Fender Apron Lower Panel	Steel
6	Fender Apron Panel Reinforcement Left Side	Steel
7	Fender Apron Panel Reinforcement Right Side	Steel
8	Fender Apron Panel	Steel
9	Secondary Bulkhead Mounting Panel	Steel
10	Front Side Member and Suspension Top Mount Assembly	Steel/High Strength Steel/Aluminum (Cast)
11	Suspension Top Mount	Aluminum (Cast)

Glazing.
Land Rover vehicles use either laminated or tempered glass for their glazing. Laminated glass is typically used for the windscreens and roof glass whilst tempered glass is used for side windows, however some vehicles can be fitted with laminated side windows as an option.
Please refer to the relevant Rescue Sheet for detailed information about a specific vehicle.

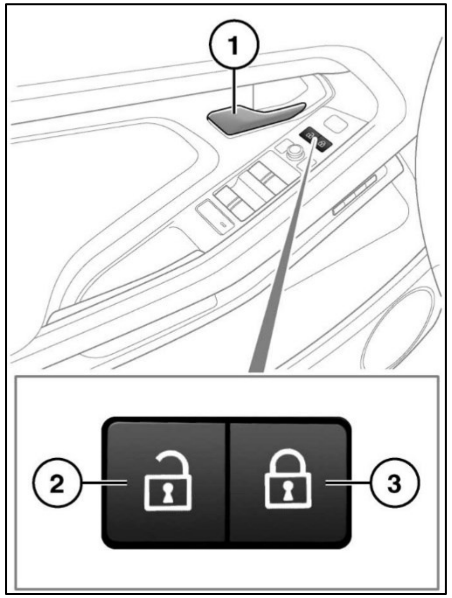
Vehicle access – Door handles.
Land Rover uses a deployable door handle design on some of their vehicles; Range Rover (L460), Range Rover Sport (L461), Range Rover Velar (L560) and Range Rover Evoque (L551). Under normal operating conditions the handles will deploy when the vehicle is unlocked, in scenarios where the system is not operational, such as loss of electrical systems, it is possible to manually operate the handle. The handles will also be deployed by the restraint system in the event of an accident.

Manual operation
1. Push the front edge of the door handle to present the rear of the handle.
2. Pull the rear of the handle until it reaches the deployed position.
3. The door can now be opened as normal.

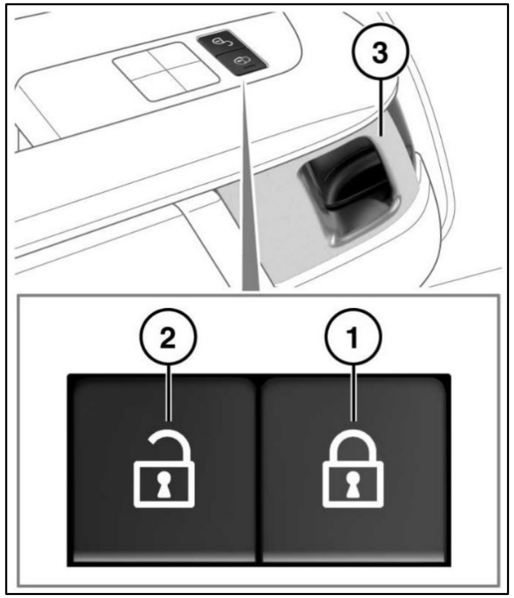
Note: This manual operation applies for all Land Rover vehicles.

Range Rover (L460) example

Vehicle access – Interior door handles.
All Land Rover vehicles feature unlocking buttons and manual release handles on the interior of the vehicle.
Operating either of the front door handles will unlock all doors, unless the vehicle has been single locked with a smart key, in this scenario only the door that has been operated will open.



Interior door handle example – Land Rover Evoque (L551)



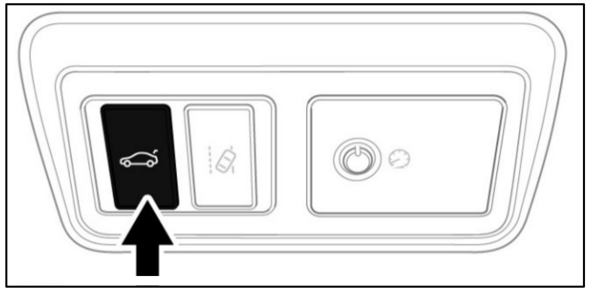
Interior door handle example – Land Rover Discovery (L462)

Vehicle access – Tailgate.
Access to the tailgate can be achieved through four different methods; Key, Interior button, and Exterior handle.

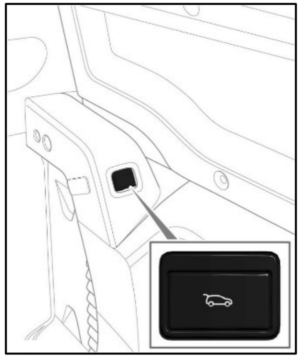
Key
Press and release the tailgate button on the key to operate the tailgate. If the vehicle is locked, the tailgate will open but all other doors will remain locked.

Key – tailgate release button

Interior button
An interior release button can be found inside the cabin, depending on vehicle and specification. The buttons will typically be found in the roof console or door.

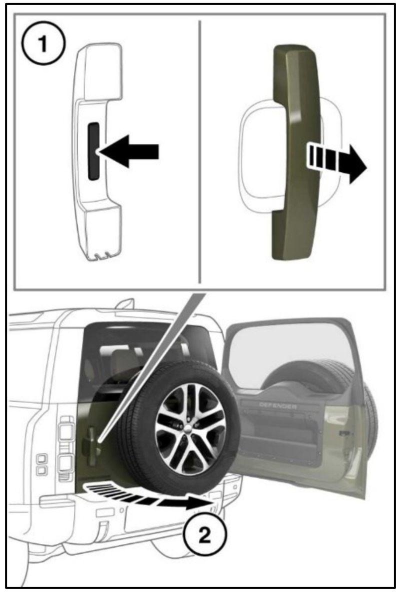


Interior button – Roof console

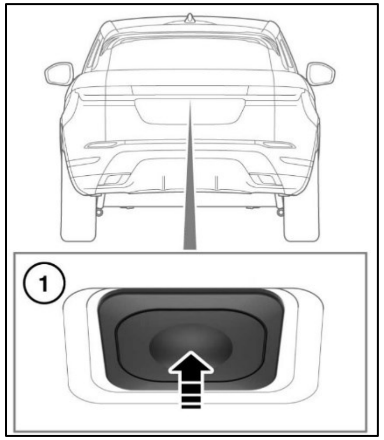


Interior button - Door

Exterior handle
Land Rovers feature an exterior switch to open the tailgate; this can either be found on the exterior handle or at the top of the number plate recess.
The tailgate button will only operate if the vehicle is unlocked or if the vehicle detects the Smart Key is present.



Exterior button – Exterior handle.



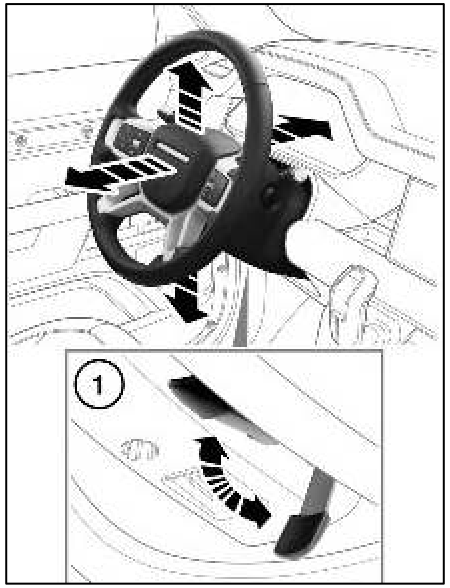
Exterior button – Number plate recess.

Steering wheel adjustment.
The adjustment of the steering wheel is dependant on the specification of the vehicle; this will either be manually or electronically.

Manual operation – A lever can be found underneath the steering wheel, once released the steering wheel can be moved.

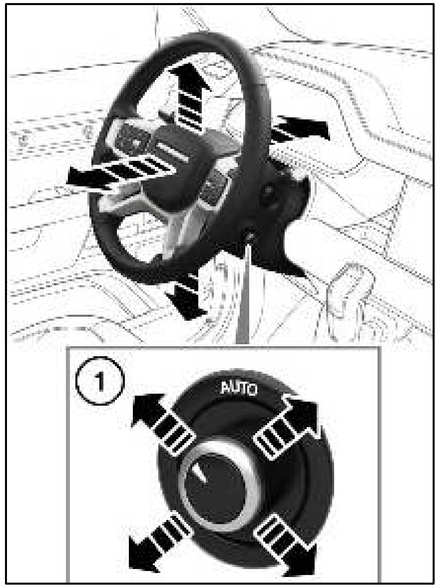
Electronic operation – A switch can be found on the side of the steering wheel, toggle the switch to move the steering wheel.

Manual operation example.



Land Rover Defender (L663)

Electronic operation example.



Land Rover Defender (L663)

Seat adjustment.
The adjustment of the seats is dependant on the specification of the vehicle; this will either be manual, electronic or a combination.

Manual operation – Levers can be found on the seat which will adjust the position as required.

1. Lever underneath the seat – This level will allow lateral movement of the seat once operated.
2. Lever on the side of the seat – This will provide vertical movement of the seat.
3. Knob – This will allow rotation of the seat back.

Manual operation example.

Range Rover Velar (L560)

Combination operation – Levers can be found on the seat which will adjust the position as required.

1. Lever underneath the seat – This level will allow lateral movement of the seat once operated.
2. Lever on the side of the seat – This will provide vertical movement of the seat.
3. Knob (electronic) – This will allow rotation of the seat back.

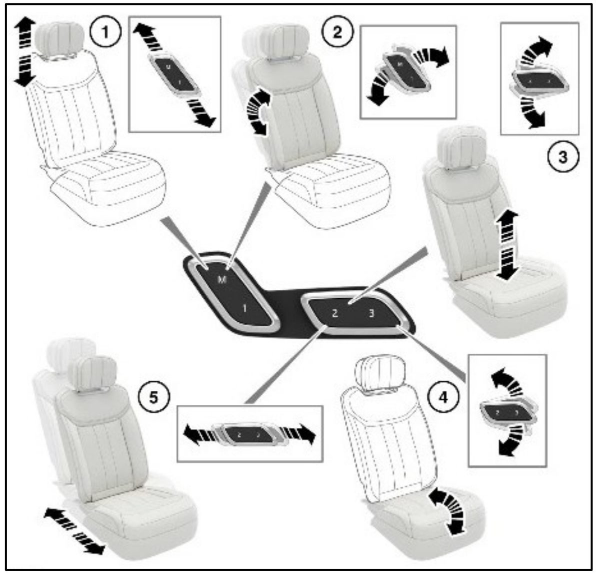
Combination operation example.

Land Rover Defender (L663)

Electronic operation – Switches will be found within the vehicle which will allow positioning of the vehicle. The location of the switches is vehicle dependant.

1. Headrest movement – Operating the highlighted switch vertically will operate the vertical position of the headrest.
2. Seat back – Rotating the highlighted switch will provide rotation of the seat back.
3. Seat height – Rotating the back of the highlighted switch will provide seat height adjustment.
4. Seat base tilt – Rotating the front of the highlighted switch will provide tilt adjustment.
5. Seat position – Laterally moving the highlighted switch will move the seat laterally.

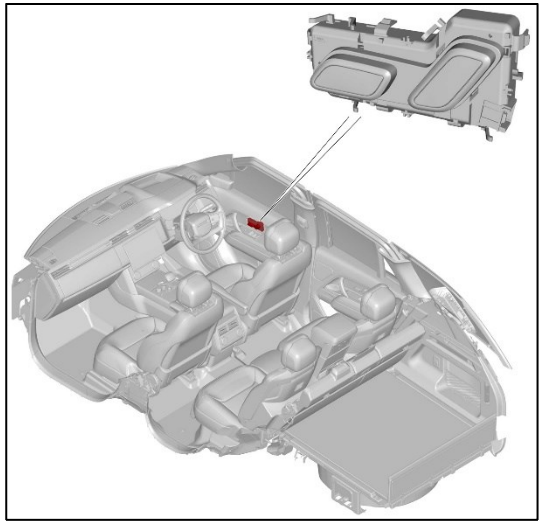
***Electronic** operation example.*



Range Rover (L460)

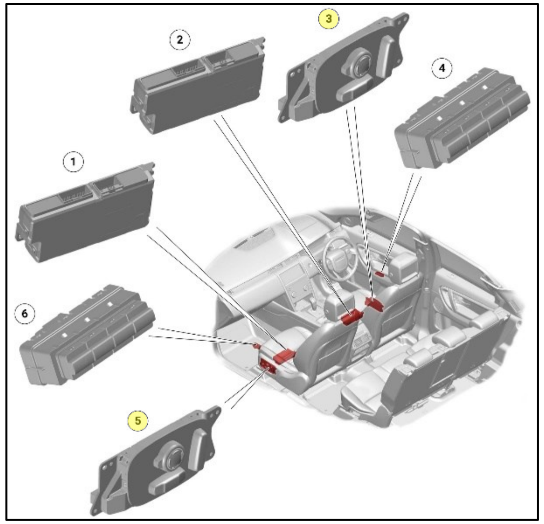
The electronic seat switches are mounted in different locations depending on the model; they will either be located on the front doors or the front seats themselves.

***Front door mounted** switch location example.*



Range Rover (L460)

***Front seat mounted** switch location example.*



Land Rover Discovery Sport (L550)

5. Stored energy / liquids / gases / solids

Stored energy – General information.

Land Rover vehicles can contain several sources and stored energy, liquid, gases, and solids. This can include items such as:

- Low and High Voltage batteries.
- Fuel tank; Petrol/Gasoline and Diesel.
- Liquid engine or battery coolant.
- Air-conditioning coolant.
- Suspension hydraulic fluid.
- Air suspension reservoirs.

The materials and chemistry used within these components, such as low and high voltage batteries, can be found within Safety Data Sheets which can be found here JLR|SDS (*www.sds.jlr.com*).

Low Voltage Batteries.

Low voltage batteries are classed as a battery with an operating voltage under 60 V DC or 30 V AC, as defined by ISO 6469.
Land Rover use two different low voltage batteries, 12 V and 48 V which use a Lead Acid (Absorbent Glass Matt (AGM)) or Lithium-Ion construction.

12 V Batteries.

All Land Rover vehicles will feature a 12 V battery; some vehicles will have an auxiliary 12V battery. The position of the auxiliary battery, if equipped, will depend on the following:

- Vehicle model.
- Drive of vehicle i.e left-hand drive or right-hand drive.

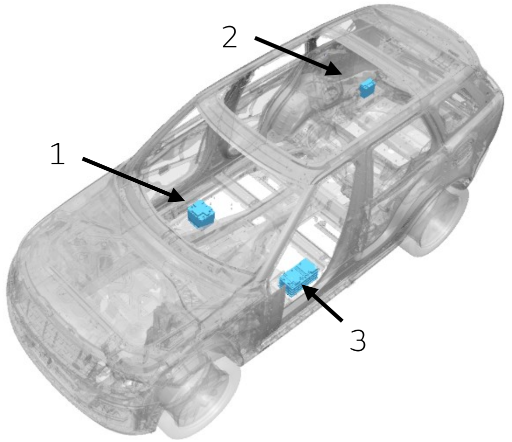
Please refer to the relevant rescue sheet for the exact information.

48 V Batteries.

Mild Hybrid Electric Vehicles (MHEV) are installed with a 48 V battery. The position of the battery will depend on the model of vehicle, please refer to the relevant Rescue Sheet for the correct location.

The locations of the low voltage batteries are shown in the following slides.

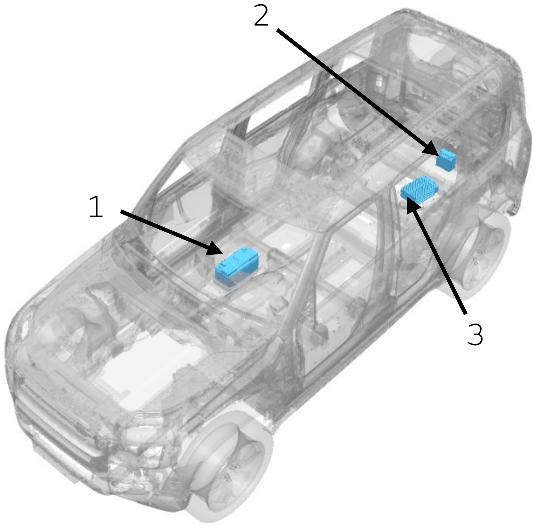
Land Rover locations of low voltage batteries



Range Rover (L460)
1. **12 v main battery (Li-ion)** – This is always located underneath the right-hand front seat.

2. **12 v auxiliary battery (Li-ion)** – This is located on the right-hand side of the loadspace; it is only fitted to PHEV vehicles.

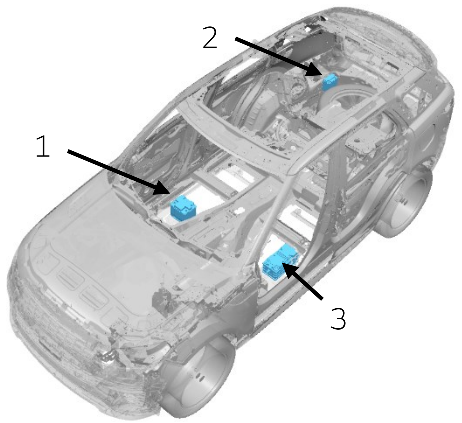
3. **48 v battery (Li-ion)** – This is located underneath the vehicle on the left-hand side; it is only fitted to MHEV vehicles.



Land Rover Defender (L663)
1. **12 v main battery** – This is located underneath the right-hand side seat on variants (90, 110 and 130). The battery material will be different for some variants:
   AGM – Non-Octa vehicles.
   Li-ion – Octa vehicles.

2. **12 v auxiliary battery (Li-ion)** – This is located underneath the loadspace floor; it is only fitted to PHEV vehicles.

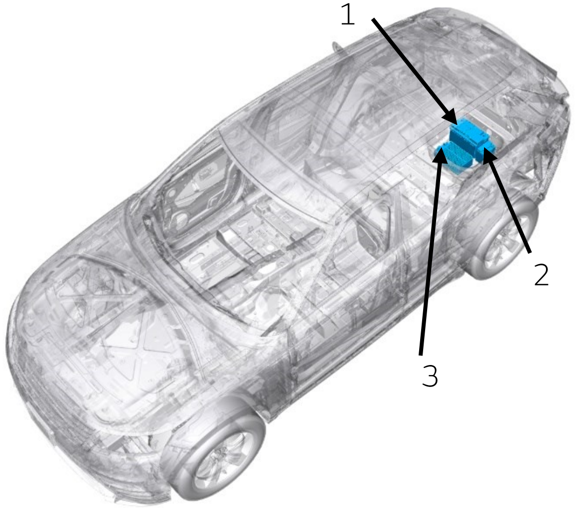
3. **48 v battery (Li-ion)** – This is located underneath the loadspace floor; it is only fitted to MHEV vehicles.



Range Rover Sport (L461)
1. **12 v main battery (Li-ion)** – This is always located underneath the right-hand front seat.

2. **12 v auxiliary battery (Li-ion)** – This is located on the right-hand side of the loadspace; it is only fitted to PHEV vehicles.

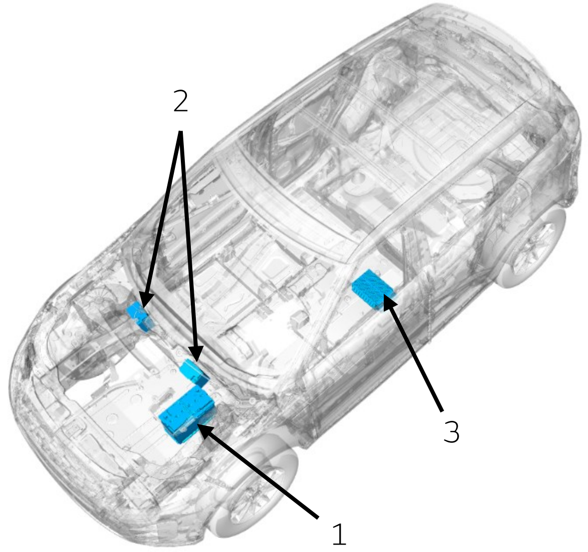
3. **48 v battery (Li-ion)** – This is located underneath the vehicle on the left-hand side; it is only fitted to MHEV vehicles.



Range Rover Velar (L560)
1. **12 v main battery (Li-ion)** – This is located underneath the loadspace floor.

2. **12 v auxiliary battery (Li-ion)** – This is located underneath the loadspace floor; it is only fitted to PHEV vehicles.

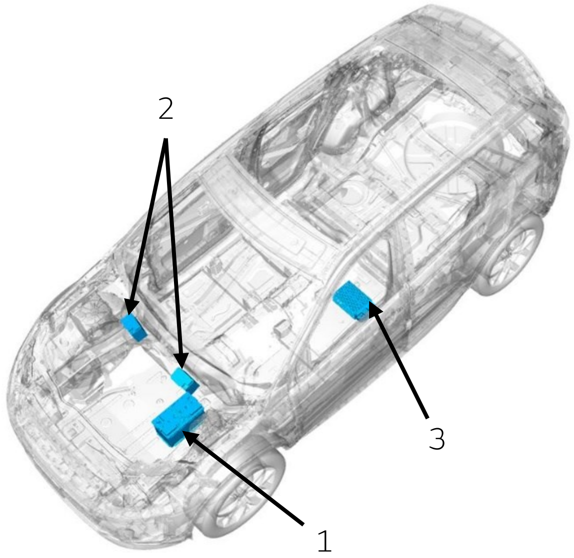
3. **48 v battery (Li-ion)** – This is located underneath the loadspace floor; it is only fitted to MHEV vehicles.



Land Rover Evoque (L551)
1. 12 v main battery (Li-ion) – This is in the engine bay.

2. 12 v auxiliary battery (Li-ion) – This is in the engine bay. The location is different for left-hand and right-hand drive vehicles. This is only fitted to PHEV vehicles.

3. 48 v battery (Li-ion) – This is located underneath the vehicle on the left-hand side; it is only fitted to MHEV vehicles.



Land Rover Discovery Sport (L550)
1. 12 v main battery (Li-ion) – This is in the engine bay.

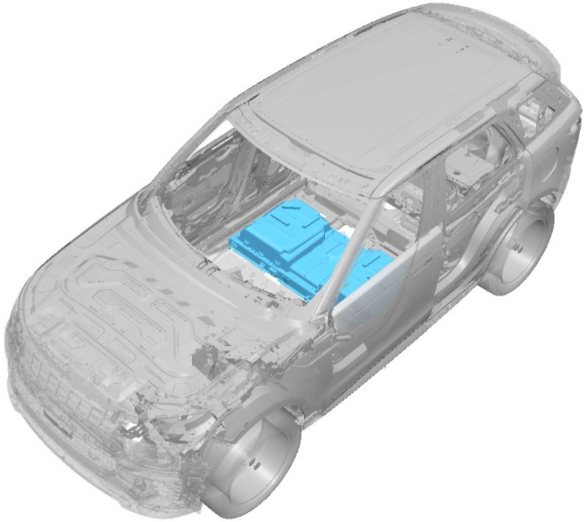
2. 12 v auxiliary battery (Li-ion) – This is in the engine bay. The location is different for left-hand and right-hand drive vehicles. This is only fitted to PHEV vehicles.

3. 48 v battery (Li-ion) – This is located underneath the vehicle on the left-hand side; it is only fitted to MHEV vehicles.

High Voltage Batteries.

High voltage batteries are classed as a battery with an operating voltage between 60 V DC and 1,500 C DV or 30 V DC and 1,000 V DC, as defined by ISO 6469.

Land Rover uses high voltage batteries as part of their Plug-In Hybrid Electric Vehicle (PHEV) powertrains, these batteries are a Lithium-Ion construction and typically operate around 400 V, please refer to Section 3 for general information on isolation or the relevant rescue sheet for vehicle specific information.

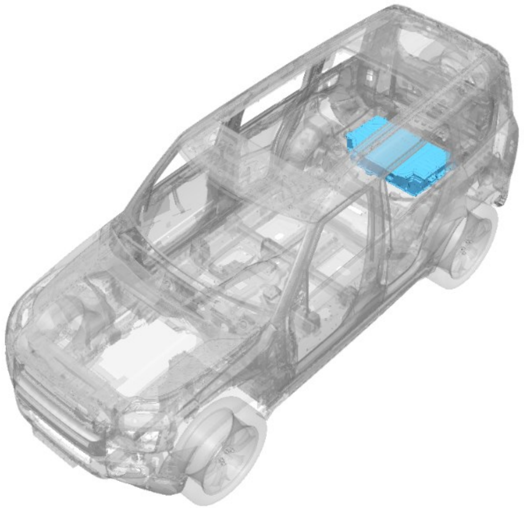


Range Rover Sport (L461)
The high voltage battery is located underneath the centre of the vehicle, in front of the fuel tank. The is no access to the battery from the inside.

Voltage – 398.2 v.
Energy storage capacity - 38.2 kWh.
Range – 91 km.
Rated capacity – 57 Ah .

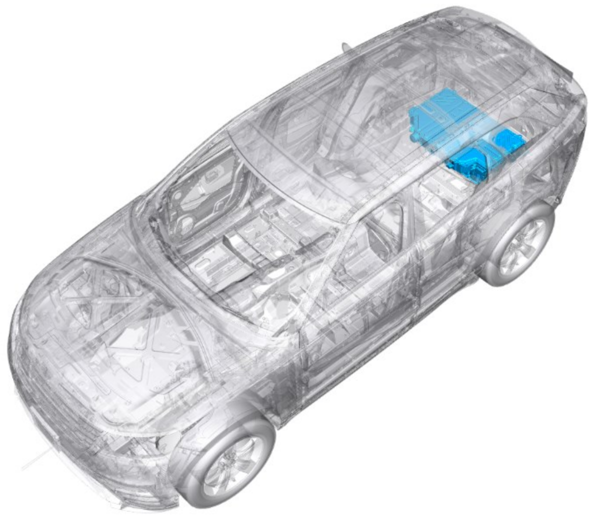
Range Rover (L460)
The high voltage battery is located underneath the centre of the vehicle, in front of the fuel tank. The is no access to the battery from the inside.

Voltage – 398.2 v.
Energy storage capacity - 38.2 kWh.
Range – 91 km.
Rated capacity – 57 Ah .



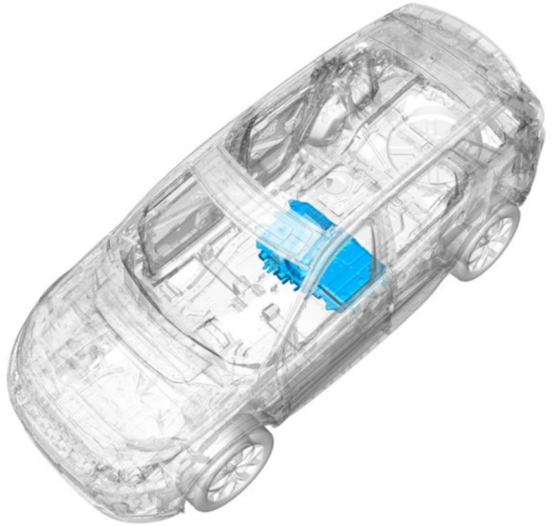
Land Rover Defender (L663)
The high voltage battery is in the loadspace of the vehicle. The loadspace floor is secured in place with fixings which need to be removed to access the battery.

Voltage – 400 v.
Energy storage capacity – 13.1 kWh.
Range - 91km.
Rated capacity – 49.5 Ah .



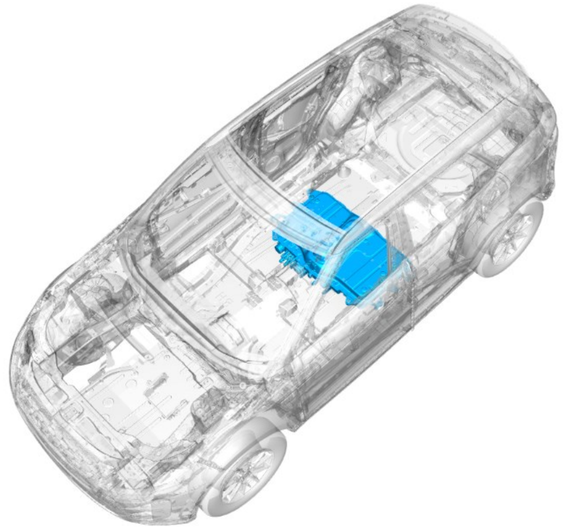
Range Rover Velar (L560)
The high voltage battery is in the loadspace of the vehicle. The loadspace floor is secured in place with fixings which need to be removed to access the battery.

Voltage – 400 v.
Energy storage capacity – 17.1 to 19.2 kWh.
Range – 53 km.
Rated capacity – 49.5 Ah .



Land Rover Discovery Sport (L550)
The high voltage battery is located underneath the centre of the vehicle, in front of the fuel tank. The is no access to the battery from the inside.

Voltage – 400 v.
Energy storage capacity – 15 kWh.
Range – 61 km.
Rated capacity – 49.5 Ah to 51 Ah



Range Rover Evoque (L551)
The high voltage battery is located underneath the centre of the vehicle, in front of the fuel tank. The is no access to the battery from the inside.

Voltage – 400 v.
Energy storage capacity – 15 kWh.
Range – 61 km.
Rated capacity – 49.5 Ah to 51 Ah.

The battery assembly cover should never be breached or removed under any circumstances, including fire. Doing so might result in severe electrical burns, shocks, or electrocution.

Fuel tanks.
All Land Rover vehicles fitted with an internal combustion engine will have a fuel tank which will contain either Gasoline or Diesel. The fuel tanks are typically located near the rear axle but please refer to the relevant Rescue Sheet for the exact location.

Liquid engine or battery coolant.
Liquid cooling systems are used to cool the engine, HV battery and some 48 V batteries.
All the coolant fluids are monoethylene glycol based which could present a risk of combustion or harm if swallowed.

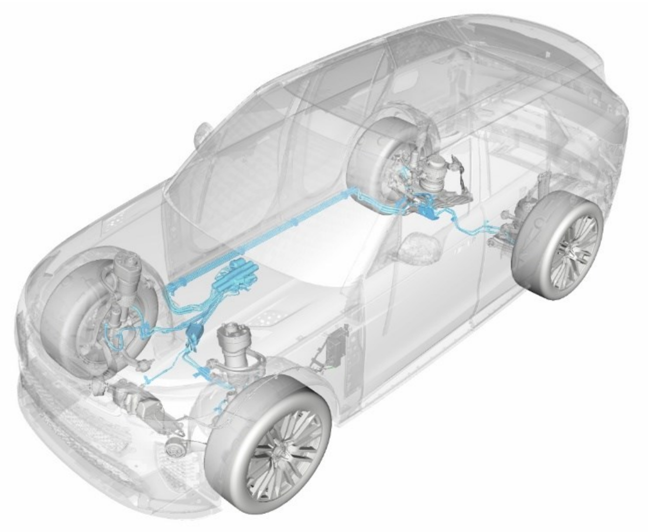
 	Extremely flammable gas. Contains gas under pressure; may explode if heated.

Air-conditioning coolant.
Air-conditioning coolant R1234y and R134a are used on Land Rover vehicles. R1234y is extremely flammable and both coolants can give off irritating or toxic fumes.

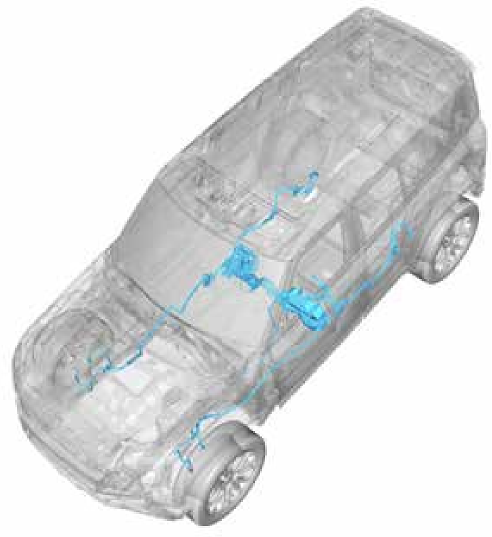
 	Extremely flammable gas. Contains gas under pressure; may explode if heated.

Suspension hydraulic fluid.
The Range Rover Sport SVR (L461) and Land Rover Defender Octa (L663) feature a unique suspension system which uses high pressure hydraulic fluid (TITAN CHF 202), operating up to 5,300 Kpa (53 bar or 767 psi).
The location of the systems are located on the relevant rescue sheets.

	Harmful if swallowed. Aspiration Hazard.



Range Rover Sport SVR (L461)



Land Rover Defender Octa (L663)

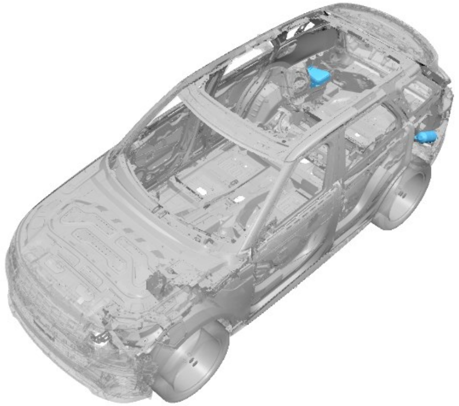
Air suspension reservoirs.

Air suspension reservoirs are compressed tanks used as part of the air suspension system. The systems can operate up to 1,800 Kpa (18 bar or 261 psi), the tanks should not be damaged or opened.

The location of the air suspension reservoirs are located on the relevant rescue sheets.

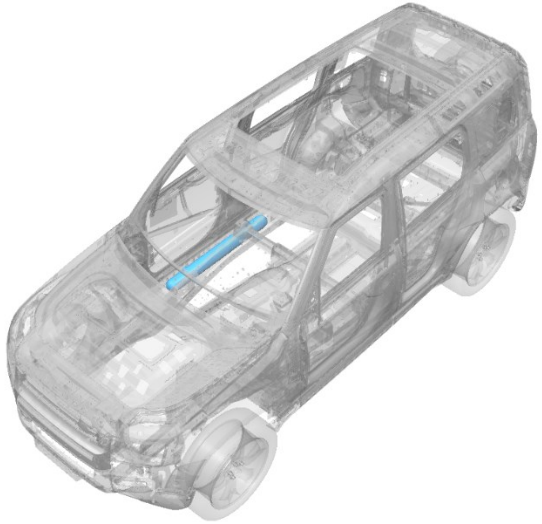
Range Rover (L460)

The air suspension compressor and reservoir tanks are located on the rear corners behind the wheel arch liners.



Range Rover Sport (L461)

The air suspension compressor and reservoir tanks are located on the rear corners behind the wheel arch liners.



Land Rover Defender (L663)

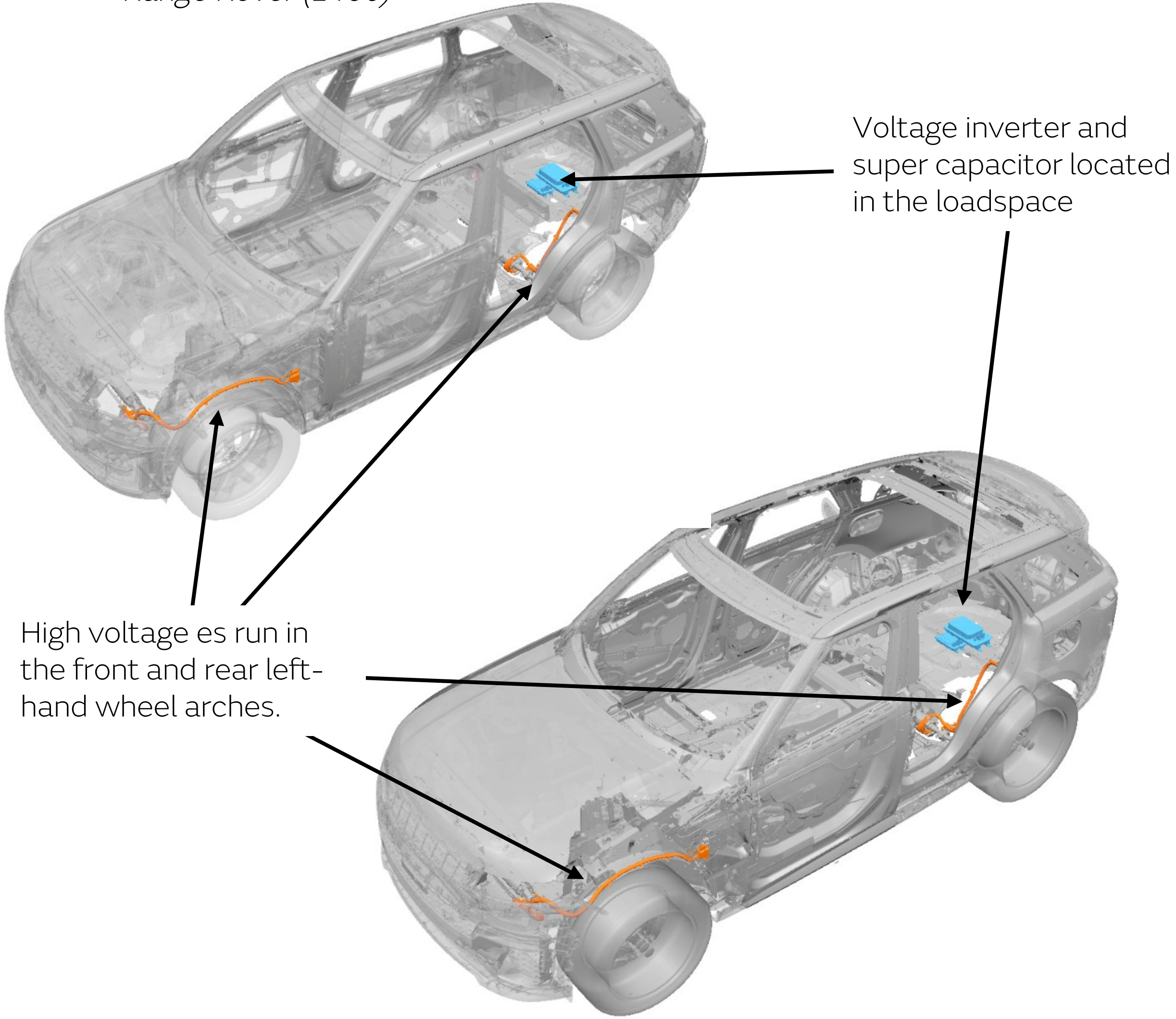
The air suspension reservoir is located on the right-hand side underneath the vehicle.

Range Rover Velar (L560)

The air suspension reservoir is located underneath the rear of the vehicle.

Suspension systems – super capacitor.
The Range Rover (L460) and Range Rover Sport (L461) vehicles have an optional suspension system which uses a super capacitor within the system. The system operates 12 V DC which is increased to 48 V DC using a power converter and stored in the super capacitor. During certain driving conditions the system can produce over 30 V RMS AC, orange high voltage ing is used in areas where this is possible. During stationary conditions, the system should not produce high voltage however caution should be taken.

Range Rover (L460)



Range Rover Sport (L461)

6. In case of fire

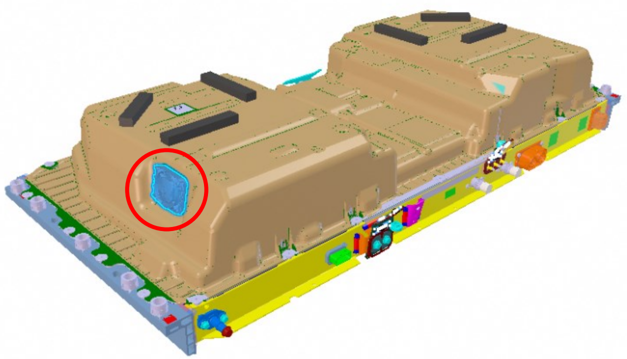
General information.
In the event of a vehicle fire, all local firefighting procedures and guidance must be followed.

> If an airbag has not deployed a thermal event may cause them to deploy, care must be taken.

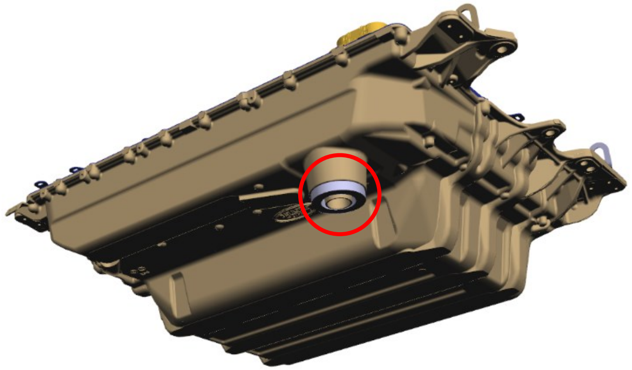
If a nonelectric vehicle is experiencing a thermal incident, the vehicle should be secured, and 12 v system should be disconnected, if possible, refer to Section 3.

High Voltage vehicles.
Vehicle fires which are not linked to the high voltage system can be extinguished with conventional methods. Care should be taken to determine if the vehicle fire is related to the High Voltage system, specifically the high voltage battery, please refer to the relevant Rescue Sheet to understand the High Voltage battery locations and layout.
PHEV and BEV vehicles can be fitted with high voltage batteries, to identify if the vehicle is fitted with a high voltage battery use the information in Section 1 to confirm the vehicle variants available or Section 5 to identify high voltage battery locations.

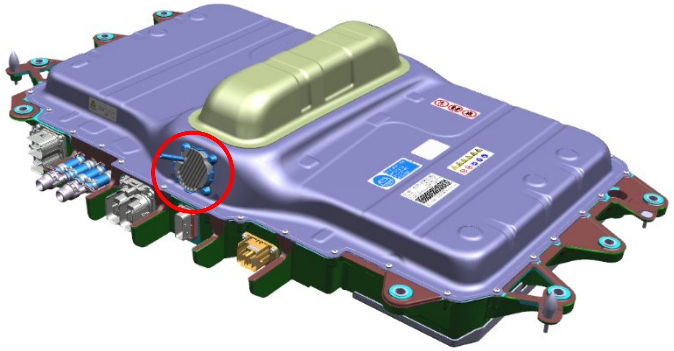
Anticipating fire in the High Voltage battery.
When assessing a fire on an electric vehicle, the high voltage components should be inspected for signs of smoke, sparks, flames, or noises which could indicate a thermal event.
High Voltage batteries are installed with vents to ventilate gases caused by internal thermal events, checking these locations will help to anticipate if there is a fire within the high voltage system.



Range Rover (L460)
and
Range Rover Sport (L461)
The high voltage battery vent is located on the right-hand side of the battery lid.



Land Rover Defender (L663)
and
Range Rover Velar (L560)
The high voltage battery vent is located on the underside of the battery housing.



Range Rover Evoque (L551)
and
Land Rover Discovery Sport (L550)
The high voltage battery vent is located on the front of the battery housing.

The temperature of the High Voltage components can also be reviewed as another indication of an internal fire or potential for a thermal incident to occur. A non-contact thermometer, such as n infrared thermometer, should be used. The temperature should be taken at several points across the high voltage component as there might be a localised internal thermal incident.
A high voltage with an increasing temperature, or exceeding 60˚c (122˚F) is an indication of an ongoing thermal event.

IR

An infrared camera should be used to detect a fire.

Fire within the High Voltage battery.
If a thermal incident is observed within the High Voltage battery plenty of water should be applied direct to the thermal incident to suppress the flames and applying cooling to the High Voltage battery.

Once a high voltage battery fire has been suppressed there is still potential of reignition. The vehicle should be monitored using an infrared camera for any signs of reignition.

A large volume of water is required to suppress and contain High Voltage battery thermal incidents.

The attending firefighters should assess the most suitable measures to use based on the incident.

Released gases.
During a thermal incident within the High Voltage battery, the chemical reaction released gases which can be toxic.
The gases present a risk of Vapour Cloud Explosion (VCE) and harm if inhaled.

The correct PPE must be worn to protect against harmful gases that could be released.

7. In case of submersion

Submerged High Voltage vehicle.

If a High Voltage vehicle is submerged, the latest industry guidelines should always be followed. Please refer to your local firefighting authority.

Appropriate PPE for water rescue should be worn.

If a High Voltage vehicle is submerged in water, it should not pose any additional risk of electric shock, although additional care should be taken as hazards may not be obvious during a visual inspection.

If the submerged vehicle is still attached to a charger, then the charger should be isolated and made safe before removing the charge lead before any recovery operation.

Following submersion, the vehicle still presents a risk of an HV electrical fire even when dried. If necessary de-activate the High Voltage Systems once the vehicle has been removed from the water.

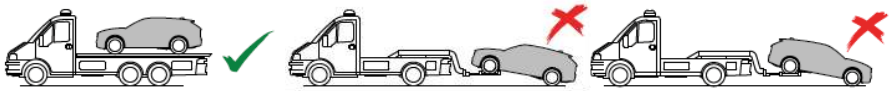
It is still possible for a battery to go into thermal runaway once removed; the vehicle and High Voltage battery should be monitored for signs of thermal runaway as discussed in Section 6.

Source: UK Department of Transport, Recovery operators: working with electric vehicles.

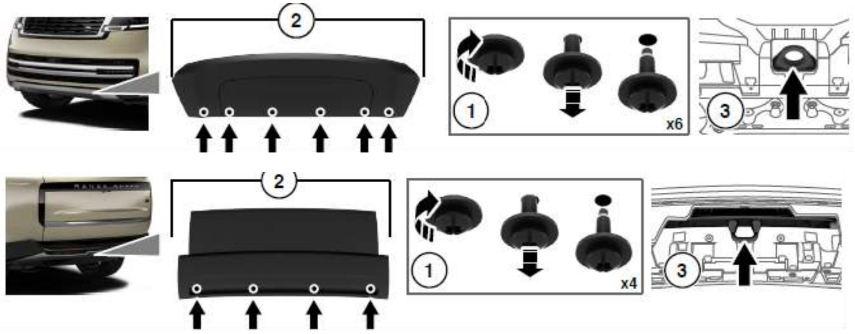
8. Towing / transport / storage

Vehicle recovery.

Flatbed is the preferred method of recovery. For High Voltage vehicles, towing could potentially damage the vehicle, and the movement can generate an electrical current which causes a risk of shock or reignition of a thermal incident.
If necessary, in an emergency, the vehicle can be moved to avoid an incident however the distance and speed should be kept to a minimum.



The vehicles should be towed using the designated location where possible. Please refer to the relevant Rescue Sheet for the position of the towing eyes.



Front and rear towing location examples, Range Rover (L460)

Care should be taken if the vehicle was involved in a thermal incident as there is a risk of reignition.

Vehicle storage.

If a High Voltage vehicle requires storage after a thermal incident, care should be taken as reignition is possible days after the initial incident.

9. Important additional information

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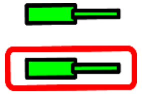
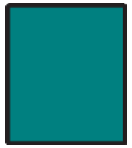
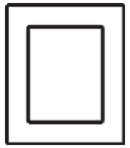
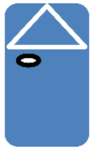
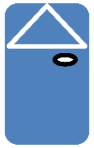
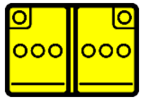
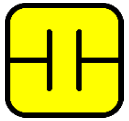
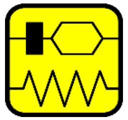
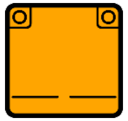
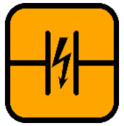
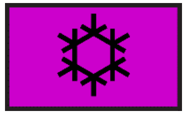
10. Explanation of pictograms used

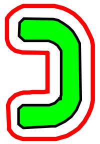
	**Airbag inflator/stored gas inflator.** To identify an airbag inflator/stored gas inflator.
	**Seat belt pretensioner.** To identify a seat belt pretensioner.
	**Gas strut / Preloaded spring.** To identify a gas strut, preloaded spring.
	**Pedestrian protection active system.** To identify the pedestrian protection active system.
	**High strength zone.** To identify a high strength zone.
	**Zone requiring special attention.** To identify the zone requiring special attention.

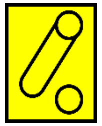
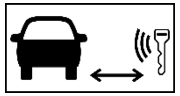
	**Carbon structure.** To indicate that carbon is used in the chassis structure. To inform about the risks of inhalation, appropriate PPE is needed.
	**Left hand drive.** To identify a left-hand drive vehicle.
	**Right hand drive.** To identify a right-hand drive vehicle.
	**Battery low voltage.** To identify a low voltage battery.
	**Ultra-capacitor, low-voltage.** To identify a low voltage ultra-capacitor.
	**SRS control unit.** To identify an SRS control unit.

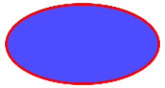
	Battery pack, high-voltage. To indicate a high voltage battery pack.
	High voltage ultra-capacitor. To indicate an ultra-capacitor pack.
	High voltage component. To indicate a high voltage component.
	High voltage power e. To identify a high voltage power e.
	Fuel tank content Diesel. To indicate the content of the tank by using a defined colour.
	Fuel tank content gasoline/ethanol. To indicate the content of the tank by using a defined colour.

	Air tank. To indicate an air tank.
	Air-conditioning component. To indicate an air conditioning component by using a defined colour.
	General warning sign. To signify a general warning.
	Warning, Electricity. To warn of electricity and dangerous voltage.
	Use thermal Infrared camera. To indicate that a thermal infrared camera should be used to detect a fire.
	Explosive. To indicate the risk of an explosion.

	Flammable. To indicate a risk of flammability.
	Gases under pressure. To indicate the risk of gases under pressure.
	Oxidizer. To indicate the risk of oxidizing material/substances.
	Corrosives. To indicate the risk of corrosive material/substances..
	Hazardous to the human health. To indicate the risk of damaging human health.
	Acute toxicity. To indicate the risk of acute toxicity.

	Environmental hazard. To indicate the risk of environmental hazard.
	Automatic rollover protection system. To indicate an automatic rollover protection component.
	Vehicle on fuel of liquid group 1. To identify vehicles using Diesel or Bio Diesel.
	Vehicle on fuel of liquid group 2. To identify vehicles using Petrol/Gasoline.
	Electric Vehicle. To identify Electric Vehicles.
	Hybrid Electric Vehicle on fuel of liquid group 2. To identify Hybrid Electric Vehicles using Petrol/Gasoline.

	**Bonnet; hood.** To identify the control that opens the compartment located outside the passenger area in the front of the vehicle. A frame may be used to separate the pictogram from the background as needed.
	**Boot; trunk.** To identify the control that opens the compartment located outside the passenger area in the rear of the vehicle. A frame may be used to separate the pictogram from the background as needed.
	**Device to shut down power in the vehicle.** Shutdown power in vehicle, in all forms, by means of: • Ignition key; • Push button; • Operation in engine compartment; • Operation on dashboard; • Battery switch; • Other.
	**Remove smart key /starter key.** To indicate that the smart key should be removed from the vehicle to prevent starting of the vehicle. A safe distance may optionally be indicated.
	**Dangerous voltage.** To indicate hazards arising from dangerous voltages. Orange = High Voltage (Class B Voltage) Yellow = Controlling the High Voltage by Low Voltage Orange frame = Procedure for disabling High Voltage vehicle
	**e cut.** To identify the e to cut that disconnect high voltage and SRS components. To show that two separate places in the same e shall be cut. Size and proportions can be adjusted to fit the intended purpose.

	**Disconnect high voltage device (e.g. service plug).** To identify the high voltage device that disconnects the high voltage, where appropriate PPE is needed for the action.
	**Disconnect high voltage device.** To identify the low voltage device that disconnects the high voltage.
	**Steering wheel, tilt control.** To identify the control that allows adjustment of the steering wheel by tilting up or down. A frame may be used to separate the pictogram from the background as needed.
	**Seat height adjustment.** To identify the control that moves the entire seat upward and downward. A frame may be used to separate the pictogram from the background as needed.
	**Seat adjustment, longitudinal.** To identify the control that moves the entire seat forward and rearward. A frame may be used to separate the pictogram from the background as needed.
	**Airbag.** To identify an airbag.